

Consumer Shopping Behaviour Analysis Project

Project Background

A national retail organisation aims to gain deeper insights into customer purchasing behaviour to enhance business performance, strengthen customer relationships, and improve retention. Recent shifts in buying habits have been observed across different customer groups, product categories, and shopping channels (online and in-store). The company wants to understand how factors such as promotional discounts, product ratings, seasonal demand, and preferred payment methods influence purchasing decisions and customer loyalty.

The primary objective of this project is to answer the following business question:

"How can consumer shopping data be analysed to identify purchasing trends, improve customer engagement, and support more effective marketing and product decisions?"

Project Deliverables

1. Data Preparation and Processing (Python)

- Import the raw dataset.
- Clean missing, duplicate, and inconsistent records.
- Transform and prepare the data for analysis using Python.

2. Database Design and Analysis (SQL)

- Create a structured relational database from the prepared dataset.
- Simulate retail transactions where required.
- Develop SQL queries to analyse customer behaviour, purchasing trends, customer loyalty, and key sales drivers.

3. Interactive Dashboard (Power BI)

- Design an interactive dashboard presenting key performance indicators and consumer insights.
- Include visualisations for customer demographics, purchasing patterns, sales performance, product categories, and shopping channels.
- Enable stakeholders to explore trends through interactive filters and slicers.

4. Project Report and Presentation

- Produce a professional report summarising the methodology, analysis, findings, and business recommendations.
- Prepare a presentation that communicates the most important insights and practical recommendations for management.

5. GitHub Repository

Create a well-organised GitHub repository containing:

- Python scripts for data preparation and transformation.
- SQL database scripts and analytical queries.
- Power BI dashboard files.
- Project documentation, including the report and presentation materials.